

DEEPAK RAKSHIT

AI Engineer Intern | Agentic AI Systems | Full-Stack Development

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EDUCATION

B.Tech in Computer Science & Engineering (AI Twin)

2nd Year (Currently Pursuing)

Noida Institute of Engineering and Technology (NIET), Greater Noida

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL, HTML/CSS

AI & Agents: Gemini 3.8 Live API, multimodal AI, agentic systems, tool calling, model routing, SQLite FTS5, memory systems

Frameworks & Backend: FastAPI, Next.js 15, React, Node.js, SQLAlchemy, Alembic, REST APIs, WebSockets, JSON-RPC 2.0

Data & DevOps: PostgreSQL, Redis, RQ, Docker, Docker Compose, Railway, Git, asynchronous workers

Security & Automation: JWT, HMAC, Razorpay, Microsoft UI Automation (COM), Playwright, WASAPI, WebRTC

PROJECTS

JARVIS | Personal AI Operating System for Windows

github.com/deepakrakshit/jarvis-ai

Python, Node.js, Gemini 3.8 Live API, Windows UI Automation, Playwright, WebSockets

- Architected a control-plane-first agent where the model only proposes tool calls and a local policy engine and capability firewall authorize every side effect.
- Built realtime voice and vision on the Gemini 3.8 Live API with WASAPI loopback capture and a PBFDFAF echo canceller, producing clean 16 kHz mono PCM with barge-in handling.
- Designed a 4-tier execution hierarchy (native URI/CLI, COM UI Automation, Playwright DOM, Node.js JSON-RPC 2.0 CUA) with closed-loop Observe, Act, Verify state confirmation.
- Shipped autonomous WhatsApp VoIP calling and a Telegram remote-control agent with cryptographic pairing and inline approvals; enforced quality with 169+ automated tests, mypy strict and ruff.

PdFORBIT | Production PDF SaaS

github.com/deepakrakshit/pdforbit

Next.js 15, FastAPI, PostgreSQL, Redis/RQ, Razorpay, Railway

- Built and deployed a full-stack PDF platform with 30+ tools across 6 categories: organize, optimize, convert, edit, security and document intelligence.
- Designed an async pipeline (upload, Redis/RQ queue, worker, status polling, HMAC-signed download) that keeps heavy PDF processing off the synchronous request path.
- Implemented JWT authentication with token rotation, rate limiting, retention cleanup, and server-side Razorpay payment, signature and webhook verification.
- Deployed separate frontend, API, worker and cleanup services on Railway with PostgreSQL, Redis and persistent storage; added integration tests for uploads, jobs, billing and cleanup.

HACKATHONS

Smart India Hackathon 2026, Team Unit8 | Problem Statement SIH26117: Sovereign On-Premise Agentic AI Workbench for Confidential Industrial Work